SQL PROJECT

datalab

Phase 1

1.1 Preview the Dataset

		N...	Sur...	D...		Gender	Sexual_Orienta...	LGBTQ
50	51	Domenic	Potts	IT	False	Male	Heterosexual	No
51	52	Hadley	Farrell	Finance	False	Female	Heterosexual	No
52	53	Audrey	Bolton	Marketing	False	Female	Heterosexual	No
53	54	Tara	Dallas	Sales	False	Male	Heterosexual	No
54	55	Mason	Farrant	HR	False	Male	Prefer not to say	Yes
55	56	Percy	Farrell	Operations	False	Male	Heterosexual	No
56	57	Liam	Santos	Sales	False	Male	Heterosexual	No
57	58	Joseph	Gosling	Finance	False	Male	Heterosexual	No
58	59	Phoebe	Knight	Marketing	False	Male	Heterosexual	No
59	60	Jackeline	Phillips	IT	False	Male	Heterosexual	No
60	61	John	Stone	Sales	False	Male	Heterosexual	No
61	62	Sienna	Edwards	Marketing	False	Female	Heterosexual	No
62	63	Marilyn	Patel	Sales	False	Female	Heterosexual	No
63	64	Taylor	Mitchell	HR	False	Female	Heterosexual	No
64	65	Mark	Baker	Finance	False	Male	Heterosexual	No

Rows: 2,222 ⚠ Truncated from 10,000 rows Expand

1.2 Total number of employees

index	employees_count
0	10000

Rows: 1 Expand

1.3 Total employee count by division

index	Division	employee_count_division
0	IT	2306
1	Sales	2245
2	Finance	1429
3	Marketing	1340
4	Operations	1227
5	HR	990
6	RD	463

Rows: 7 Expand

1.4 Overall gender distribution

ind...	Gender	employee_count_gender
0	Male	5902
1	Female	3388
2	Prefer not to say	397
3	Non-binary/non-conforming	110
4	Other	106
5	Transgender	97

Rows: 6 Expand

1.5 Overall ethnicity breakdown

index	Ethnicity	employee_count_ethnicity
0	White	7092
1	Asian	1718
2	Black	416
3	Middle Eastern	352
4	Latin	209
5	Prefer not to say	108
6	Other	105

Rows: 7 Expand

1.6 Count of employees with disability

index	Disability	employee_count_disability
0	No	
1	Yes	
2	Prefer not to say	

Rows: 3 Expand

1.7 Count of employees who identify as LGBTQ

index	LGBTQ	employee_count_LGBTQ
0	No	
1	Yes	
2	Prefer not to say	

Rows: 3 Expand

1.8 Count of employees who are veterans

index	Veteran	employee_count_veterans
0	No	
1	Prefer not to say	
2	Yes	

Rows: 3 Expand

1.9 Count of employees who identify as minority

index	Minority	employee_count
0	No	
1	Yes	
2	Prefer not to say	

Rows: 3 Expand

1.10 Average age by division

index	Division	employee_count
0	IT	
1	HR	
2	Sales	
3	Finance	
4	Operations	
5	Marketing	
6	RD	

Rows: 7 Expand

1.11 Employees count who don't want to disclose their gender or ethnicity

index	employee_count
0	

Rows: 1 Expand

1.12 How many unique nationalities exist in the company?

index	employee_count
0	

Rows: 1 Expand

Phase 2

2.1 Gender breakdown per division

index	Division	Gender	employee_count
0	IT	Male	
1	Sales	Male	
2	Finance	Male	
3	Marketing	Male	
4	Sales	Female	
5	IT	Female	
6	Operations	Male	
7	HR	Male	
8	Finance	Female	
9	Marketing	Female	
10	Operations	Female	
11	HR	Female	
12	RD	Male	
13	RD	Female	
14	Sales	Prefer not to say	
15	IT	Prefer not to say	

Rows: 42 Expand

2.2 Ethnicity breadown per division

	D...	Ethnicity	employe...
0	IT	White	1631
1	Sales	White	1614
2	Finance	White	994
3	Marketing	White	933
4	Operations	White	886
5	HR	White	694
6	IT	Asian	411
7	Sales	Asian	371
8	RD	White	340
9	Finance	Asian	258
10	Marketing	Asian	237
11	HR	Asian	189
12	Operations	Asian	184
13	IT	Black	93
14	Sales	Black	91
15	Sales	Middle Eastern	84

Rows: 49 Expand

2.3 Disability representation per division

	D...	D...	employe...
0	Yes	IT	376
1	Yes	Sales	372
2	Yes	Finance	206
3	Yes	Operations	191
4	Yes	Marketing	190
5	Yes	HR	153
6	Yes	RD	57

Rows: 7 Expand

2.4 Age group distribution using CASE WHEN

	a.	employe...
0	30 to 45	6066
1	Under 30	1990
2	Over 45	1944

Rows: 3 Expand

2.5 Age group distribution by division

	D...	a.	employe...
0	Finance	30 to 45	873
1	Finance	Under 30	288
2	Finance	Over 45	268
3	HR	30 to 45	624
4	HR	Over 45	191
5	HR	Under 30	175
6	IT	30 to 45	1399
7	IT	Over 45	462
8	IT	Under 30	445
9	Marketing	30 to 45	796
10	Marketing	Under 30	281
11	Marketing	Over 45	263
12	Operations	30 to 45	718
13	Operations	Under 30	269
14	Operations	Over 45	240
15	RD	30 to 45	274

Rows: 21 Expand

2.6 Number of unique nationalities per division

	D...	unique_nationalities
0	Sales	104
1	IT	102
2	Finance	93
3	Marketing	87
4	Operations	86
5	HR	83
6	RD	56

Rows: 7 Expand

2.7 LGBTQ representation by division

	D...	lgbtq_em...
0	IT	282
1	Sales	255
2	Finance	162
3	Marketing	158
4	Operations	143
5	HR	117
6	RD	58

Rows: 7 Expand

2.8 Division with the most underrepresented employees

(counts employees who fall into at least one group: Minority, Disability, LGBTQ, Indigenous, Veteran)

	D...	underrepresented_employees
0	IT	1045
1	Sales	1025
2	Finance	646
3	Marketing	594
4	Operations	536
5	HR	444
6	RD	203

Rows: 7 Expand

2.9 Gender distribution - only divisions with more than 500 employees

	D...		employe...
0	IT	Male	1384
1	Sales	Male	1323
2	Finance	Male	845
3	Marketing	Male	782
4	Sales	Female	776
5	IT	Female	763
6	Operations	Male	723
7	HR	Male	585

Rows: 8 Expand

2.10 Most common etnicity per division

	D...	Ethnicity	employe...
0	IT	White	1631
1	Sales	White	1614
2	Finance	White	994
3	Marketing	White	933
4	Operations	White	886
5	HR	White	694
6	IT	Asian	411
7	Sales	Asian	371
8	RD	White	340
9	Finance	Asian	258
10	Marketing	Asian	237
11	HR	Asian	189
12	Operations	Asian	184
13	IT	Black	93
14	Sales	Black	91
15	Sales	Middle Eastern	84

Rows: 49 Expand

Phase 3

3.1 Importing a new data for Joining phase

	D...	L.	Department_...	T.	Annual_Budg...	DEI_Bud...	Remot...	Foun...
0	IT	Dubai	Ahmed Al Mansoori	2306	4500000	120000	Hybrid	
1	Sales	Abu Dhabi	Sarah Johnson	2245	3800000	95000	On-site	
2	Finance	Riyadh	Mohammed Al Farsi	1429	3200000	80000	Hybrid	
3	Marketing	Dubai	Priya Sharma	1340	2800000	75000	Remote	
4	Operations	Doha	James Williams	1227	2600000	60000	On-site	
5	HR	Abu Dhabi	Fatima Al Zaabi	990	1800000	95000	Hybrid	
6	RD	Dubai	Chen Wei	463	5200000	110000	Remote	

Rows: 7 Expand

3.2 Merging and reviewing the table

				D		Ge...	Sexual_Orienta...	LGBTQ	I...
0	1	Deborah	Addison	IT	No	Transgender	Heterosexual	Prefer not to say	No
1	2	Mona	Hill	Marketing	No	Male	Heterosexual	No	Yes
2	3	Kimberly	Shelton	Finance	No	Female	Heterosexual	No	No
3	4	Denis	Robinson	HR	No	Male	Heterosexual	No	No
4	5	Carmen	Gunn	RD	No	Male	Heterosexual	No	No
5	6	Nathan	Furnell	Sales	No	Male	Heterosexual	No	No
6	7	Luke	Sinclair	Marketing	No	Male	Prefer not to say	Yes	No
7	8	Fred	Callan	Sales	No	Male	Heterosexual	No	No
8	9	Carter	Owen	Finance	No	Male	Heterosexual	No	No
9	10	Harry	Dobson	Sales	No	Male	Heterosexual	No	No

Rows: 10 Expand

3.3 Total number of employees in merged tables

	employee...
0	10000

Rows: 1 Expand

3.4 Checked all the divisions and their locations

	D...	L.	Department_...	Remot...
0	Finance	Riyadh	Mohammed Al Farsi	Hybrid
1	HR	Abu Dhabi	Fatima Al Zaabi	Hybrid
2	IT	Dubai	Ahmed Al Mansoori	Hybrid
3	Marketing	Dubai	Priya Sharma	Remote
4	Operations	Doha	James Williams	On-site
5	RD	Dubai	Chen Wei	Remote
6	Sales	Abu Dhabi	Sarah Johnson	On-site

Rows: 7 Expand

3.5 Average diversity score by gender

	Gender	avg_diversity_positive	avg_diversity_negative
0	Other	2.75	1.39
1	Male	2.67	1.55
2	Prefer not to say	2.66	1.6
3	Non-binary/non-conforming	2.65	1.74
4	Female	2.63	1.56
5	Transgender	2.39	1.77

Rows: 6 Expand

3.6 Average equity score by ethnicity

	Ethnicity	avg_equity_posit...	avg_equity_nega...
0	White	3.11	1.25
1	Prefer not to say	3.11	1.21
2	Other	3.08	1.45
3	Asian	3.07	1.29
4	Middle Eastern	3.04	1.29
5	Latin	3	1.26
6	Black	2.99	1.3

Rows: 7 Expand

3.7 Average inclusion score by division (with location)

	D...	L.	avg_inclusion_positive	avg_inclusion_negative
0	RD	Dubai	3.01	1.27
1	Sales	Abu Dhabi	3	1.34
2	Finance	Riyadh	2.97	1.36
3	HR	Abu Dhabi	2.96	1.37
4	IT	Dubai	2.96	1.35
5	Operations	Doha	2.92	1.39
6	Marketing	Dubai	2.91	1.43

Rows: 7 Expand

3.8 Negative sentiment count by ethnicity

	Ethnicity	total_diversity_negative	total_equity_negative	total_inclusion_negative
0	White	10831	8864	9564
1	Asian	2788	2219	2414
2	Black	662	539	581
3	Middle Eastern	596	454	479
4	Latin	321	263	298
5	Other	177	152	153
6	Prefer not to say	181	131	149

Rows: 7 Expand

3.9 Which gender has the highest negative inclusion responses?

	Gender	total_inclusion_negative
0	Male	7937
1	Female	4721
2	Prefer not to say	529
3	Transgender	163
4	Non-binary/non-conforming	163
5	Other	125

Rows: 6 Expand

3.10 DEI budget vs average inclusion score per division

	D...	DEI_Bud...	Remot...	avg_inclusion_sc...
0	IT	120000	Hybrid	2.96
1	RD	110000	Remote	3.01
2	Sales	95000	On-site	3
3	HR	95000	Hybrid	2.96
4	Finance	80000	Hybrid	2.97
5	Marketing	75000	Remote	2.91
6	Operations	60000	On-site	2.92

Rows: 7 Expand

3.11 Remote policy vs diversity sentiment

	Remot...	avg_diversity_positive	avg_diversity_negative	employe...
0	Remote	2.67	1.53	1803
1	Hybrid	2.66	1.56	4725
2	On-site	2.65	1.57	3472

Rows: 3 Expand

3.12 Overall DEI score by division with full context

	D...	L.	Remot...	DEI_Bud...	avg_di...	a...	avg_in...
0	RD	Dubai	Remote	110000	2.79	3.11	3.01
1	Sales	Abu Dhabi	On-site	95000	2.66	3.08	3
2	Finance	Riyadh	Hybrid	80000	2.67	3.1	2.97
3	IT	Dubai	Hybrid	120000	2.65	3.09	2.96
4	HR	Abu Dhabi	Hybrid	95000	2.65	3.12	2.96
5	Operations	Doha	On-site	60000	2.62	3.07	2.92
6	Marketing	Dubai	Remote	75000	2.63	3.11	2.91

Rows: 7 Expand

Phase 4

Part A: CASE, SUBQUERIES

4.1 Composite DEI sentiment label per employee using CASE WHEN

	Gender	Ethnicity	D...	total_po...	total_ne...	overall_dei_sentiment
0	Transgender	White	IT	5	6	Negative
1	Male	White	Marketing	8	2	Positive
2	Female	White	Finance	8	5	Positive
3	Male	White	HR	7	3	Positive
4	Male	Asian	RD	11	4	Positive
5	Male	Black	Sales	10	5	Positive
6	Male	White	Marketing	10	3	Positive
7	Male	White	Sales	9	5	Positive
8	Male	Black	Finance	12	2	Positive
9	Male	Black	Sales	10	4	Positive
10	Male	White	Sales	9	3	Positive
11	Female	Middle Eastern	RD	7	7	Neutral
12	Prefer not to say	Black	Operations	10	4	Positive
13	Male	Middle Eastern	Operations	10	3	Positive
14	Male	White	Sales	6	7	Negative
15	Male	Middle Eastern	IT	11	3	Positive

Rows: 25 Expand

4.2 Division where minority representation is below company average

	D...	minority...
0	Marketing	286
1	RD	107
2	Operations	255
3	HR	230

Rows: 4 Expand

4.3 Most at-risk group-lowest inclusion sentiment

	Ge...	inclusion_...
0	Transgender	0.85

Rows: 1 Expand

4.4 DEI sentiment breakdown by ethnicity using CASE WHEN

	Ethnicity	total_emp...	positive_sentim...	negative_senti...
0	White	7092	6095	655
1	Asian	1718	1402	210
2	Black	416	335	50
3	Middle Eastern	352	296	37
4	Latin	209	171	22
5	Other	105	79	16
6	Prefer not to say	108	90	13

Rows: 7 Expand

4.5 Final: 5 key D&I findings in one query

	D...	total_emp...	fema...	minority...	disability_c...	avg_inclusion_sc...
0	RD	463	172	107	57	1.74
1	Sales	2245	776	503	372	1.66
2	IT	2306	763	498	376	1.61
3	Finance	1429	473	327	206	1.61
4	HR	990	337	230	153	1.59
5	Operations	1227	403	255	191	1.53
6	Marketing	1340	464	286	190	1.48

Rows: 7 Expand

4.6 Divisions where female employee count is BELOW company average

	D...	fema...
0	RD	172
1	HR	337
2	Operations	403
3	Marketing	464
4	Finance	473

Rows: 5 Expand

4.7 Employees whose age is above the overall company average age

	D...	Gender	Ethnicity	
0	Sales	Female	White	65
1	HR	Male	Black	65
2	RD	Female	White	65
3	Sales	Male	Asian	65
4	RD	Female	Asian	65
5	Sales	Prefer not to say	White	65
6	Sales	Male	White	65
7	Sales	Male	White	65
8	IT	Male	White	65
9	IT	Male	White	65
10	Marketing	Male	White	65

Rows: 4,565 Expand

4.8 Division where minority representation is BELOW company average

	D...	minority...
0	RD	107
1	HR	230
2	Operations	255
3	Marketing	286

Rows: 4 Expand

4.8 Employees with inclusion score BELOW the company average

	D...	Gender	Ethnicity	inclusi...
0	Marketing	Female	Prefer not to say	5
1	IT	Prefer not to say	Asian	5
2	Marketing	Transgender	White	5
3	Sales	Male	White	5
4	Finance	Female	Other	5
5	Operations	Male	White	5
6	Sales	Male	White	5
7	Finance	Female	White	5
8	IT	Female	White	5
9	Operations	Male	Asian	5
10	Finance	Male	White	5

Rows: 20 Expand

Part B: CTEs (Common Table Expressions)

4.9 CTE- Composite DEI score per division

	D...	diversity_...	equit...	inclusion_...	composite_dei_s...
0	RD	1.36	1.85	1.74	4.95
1	Sales	1.11	1.8	1.66	4.57
2	HR	1.1	1.87	1.59	4.56
3	IT	1.09	1.84	1.61	4.55
4	Finance	1.11	1.83	1.61	4.55
5	Marketing	1.07	1.84	1.48	4.39
6	Operations	1.02	1.81	1.53	4.37

Rows: 7 Expand

4.10 CTE- Identify the most at-risk demographic group

	Ethnicity	avg_equity...	avg_inclusion_sc...	combined_risk_s...
0	Other	1.63	1.47	3.1
1	Black	1.7	1.45	3.15
2	Latin	1.74	1.41	3.15
3	Asian	1.77	1.46	3.24
4	Middle Eastern	1.75	1.57	3.32
5	Prefer not to say	1.9	1.56	3.46
6	White	1.86	1.65	3.51

Rows: 7 Expand

4.11 CTE- Gender sentiment comparison

	Gender	employe...	diversity_...	equit...	inclusion_...
0	Transgender	97	0.62	1.21	0.85
1	Non-binary/non-conforming	110	0.91	1.26	1.17
2	Female	3388	1.07	1.23	1.54
3	Male	5902	1.13	2.24	1.64
4	Prefer not to say	397	1.06	1.38	1.67
5	Other	106	1.37	1.13	1.77

Rows: 6 Expand

4.12 CTE- Underrepresented group sentiment

	group_type	avg_inclusion_sc...	avg_equity...
0	Underrepresented	1.52	1.59
1	General Workforce	1.67	2.06

Rows: 2 Expand

Part C: Window Function

4.13 Rank divisions by composite DEI score (best to worst)

	D...	diversity_...	equit...	inclusion_...	composite...	
0	RD	1.36	1.85	1.74	4.95	1
1	Sales	1.11	1.8	1.66	4.57	2
2	HR	1.1	1.87	1.59	4.56	3
3	IT	1.09	1.84	1.61	4.55	4
4	Finance	1.11	1.83	1.61	4.55	5
5	Marketing	1.07	1.84	1.48	4.39	6
6	Operations	1.02	1.81	1.53	4.37	7

Rows: 7 Expand

4.14 Running total of underrepresented employees by Division

	D...	underrepresented_in_division	runnin...
0	Finance	598	598
1	HR	416	1014
2	IT	973	1987
3	Marketing	539	2526
4	Operations	497	3023
5	RD	189	3212
6	Sales	952	4164

Rows: 7 Expand

4.15 Rank ethnicities by average inclusion score within each division

	D...	Ethnicity	avg_inclusion_sc...	rank_within_division
0	Finance	Latin	0.42	7
1	Finance	Black	1.47	6
2	Finance	Other	1.47	5
3	Finance	Asian	1.56	4
4	Finance	Middle Eastern	1.6	3
5	Finance	White	1.66	2
6	Finance	Prefer not to say	2	1
7	HR	Asian	1.1	7
8	HR	Middle Eastern	1.53	6
9	HR	White	1.68	5
10	HR	Black	1.81	4

Rows: 49 Expand

4.16 Compare each division's DEI score to the company average

	D...	division_dei_sc...	company_av...
0	RD	4.95	4.56
1	Sales	4.57	4.56
2	HR	4.56	4.56
3	IT	4.55	4.56
4	Finance	4.55	4.56
5	Marketing	4.39	4.56
6	Operations	4.37	4.56

Rows: 7 Expand

Final Deliverable- 7 Key D&I Findings

Finding 1: Which division is most diverse by ethnicity?

		unique_ethnicit...
0	IT	7

Rows: 1 Expand

Finding 2: Which gender feels least included?

	Ge...	avg_in...
0	Transgender	0.85

Rows: 1 Expand

Finding 3: Which ethnicity feels least fairly treated (equity)?

	E	a...
0	Other	1.63

Rows: 1 Expand

Finding 4: Which Division has the lowest DEI composite score?

	D...	composite...
0	Operations	4.37

Rows: 1 Expand

Finding 5: Do underrepresented employees feel less included than others?

	group_type	avg_in...
0	Underrepresented	1.52
1	General Workforce	1.65

Rows: 2 Expand

Finding 6: Which age group has the most negative diversity sentiment?

	a.	avg_diversity_sc...
0	Under 30	1.05
1	Over 45	1.08
2	30 to 45	1.13

Rows: 3 Expand

Finding 7: Overall company DEI health summary

	company_diversity_score	company_equity_s...	company_inclusion_score	overall_dei_s...
0	1.1	1.83	1.6	4.53

Rows: 1 Expand